DOMAIN GUIDE

Bitcoin Authorization And Recovery

This domain preserves participant control of Bitcoin collateral while making fixed, previously authorized transactions operable across interruptions. A **prefund** stages borrower Bitcoin before activation; an **escrow** is the later confirmed collateral output. A **PSBT** (partially signed Bitcoin transaction) carries a transaction and signing material for review and signature. A signed package is evidence of exact prior authorization, not permission for an application to choose a new destination. Bitcoin consensus, including confirmations and timelocks, remains the authority for spendability and the final outcome.

> This is the domain guide for Bitcoin Authorization And Recovery, one of the five domains of Bitcoin-backed USDC one-loan system. The architecture guide, a separate document, explains how the domains work together.

The secrets serve different actions: borrower `X` enables acceptance and fixed activation, lender `Y` enables normal borrower recovery after repayment claim, and optional independent `S` enables the same recovery transaction under the separate finalized-refund fallback. A long-stop is instead a participant-authorized absolute-timelock route without either recovery secret. None grants the organization a collateral key.

Construction And The Two Participants

Bitcoin prefund, escrow, and package construction constructs and validates scripts, outpoints, amounts, fixed outputs, fee candidates, manifests, signatures, and the package graph. It operates only the route justified by current chain and package evidence. It consumes committed terms and acknowledged participant allocations; it cannot allocate a key or invent a missing binding. It also owns constrained fee selection, relay, independent chain observation/reconciliation, and the boundary around a separately funded fee-bump child.

Borrower Bitcoin authorization and recovery keeps the borrower Vault, seed, loan keys, `X`, and signing local to the dedicated Vault origin. It creates or restores the loan's immutable allocation, acknowledges the encrypted snapshot, prepares public destination/commitment material, and signs an exact canonical batch. Acceptance first reviews nonsecret facts; a narrowly unlocked worker prepares the exact secret-bearing transaction; a second confirmation tied to those bytes authorizes signing and submission. Password or login recovery cannot recreate a forgotten Vault password. **Lender Bitcoin authorization and recovery** applies the corresponding local boundary to the lender's key and `Y`, fixed default-package signing, normal release, manual export, and fresh cooperative signatures. Account access never exposes either participant's plaintext authority. The Vault envelope and local handoff, the participant-local Base calls, and the cooperative settlement package contracts retain these exact interactions.

Package Lifecycle And Recovery

Origination can construct and sign against an exactly observed zero-confirmation prefund after its committed binding and preparation prerequisites, but six unchanged confirmations are required for funding-window opening, the pre-X gate, and reliance. The initial family contains six activation and six borrower refund candidates at 10, 20, 50, 100, 150, and 200 sat/vB (satoshis per virtual byte). Terminal packages are constructed only for the actual activation winner after six confirmations. The package contract specifies six borrower recovery, six lender recovery, and six borrower long-stop variants, while the construction responsibility's archive-readiness rule and the durable signed exit package archive constraint describe a twelve-package signed terminal phase. This source conflict is unresolved and remains for a separate repair; neither count is chosen until it is answered. The prefund and origination attempt states, the package graph states, the origination binding and package manifest contract, and the zero-confirmation boundary remain the exact owners.

The order is unsigned construction, immutable canonical signing batch, participant Vault signing, Core cryptographic validation and complete writes to the R2 package archive, then the receipt committed in the D1 workspace database. The receipt cannot be an input to the signing action that produces it. Incomplete terminal signing after activation leaves an explicit pending condition and continued recovery work; it cannot undo principal or escrow. A different activation winner requires fresh terminal construction/signing. A changed or lost parent recursively invalidates descendants; a reorg of the same unchanged parent preserves bytes and signatures but suspends qualification. This avoids treating stale transactions as executable.

Finalized public Y, or valid independently published S for a production-activated commitment, fills the same borrower-signed recovery transaction (route code 3) without changing its output, signature, ID, leaf, or manifest entry. Lender default and borrower long-stop have their own consensus deadlines and remain concurrent once both are valid; only one confirmed spend wins. Whether every loan has the borrower long-stop route is an open question: a build constraint requires every newly constructed escrow to include long-stop candidates that need neither Y nor S, and the default timing invariant and long-stop recovery flow define it, while the missing-Y security boundary says a loan without the production-activated S commitment has no borrower fallback Bitcoin route; the records do not settle whether these describe different loans or contradict each other. A Base repayment or refund cannot revoke those signatures. Cooperative signing requires fresh participant agreement and both matching loan keys, while its complete product/accounting operation remains deferred. The finalized secret to fixed Bitcoin action contract, the escrow and terminal outcome states, the one terminal escrow outcome invariant, and the missing-Y residual lender risk preserve the exact boundary.

Fees, Failures, And Recorded Choices

Fixed fees are necessary because the parent is signed before use. Provider-neutral integer units, actual virtual size, strict buffer comparison, estimate freshness, and replacement rules prevent underestimation or unsafe reselection. The existing fee sources disagree about the case where no signed tier qualifies: the payload description and two ticket resolutions select the largest tier with an alert, while the action contract says fail closed, as an earlier explanation says a former

decision did; the records could not locate that decision. The records preserve both statements and identify the conflict; neither interpretation is authorized until it is answered. The fee estimate and signed tier selection contract holds the exact source text.

A pay-to-anchor (P2A) output permits a separately funded child-pays-for-parent (CPFP) child. Product export, deterministic unsigned construction, Core's manually unlocked signing/finalization/broadcast, and independent observation stay separate so a fee wallet gains no participant authority. The CPFP request, local helper, and Bitcoin Core contract owns the arithmetic and operator process. A relay timeout is ambiguous, not permission to send a different transaction; durable exact-attempt reconciliation and independent source agreement govern retry. Wrong binding, stale bytes, contested facts, missing primary archive evidence, or invalid route blocks the affected action.

The recorded rationale for exact scripts and witnesses is that a mismatch can strand collateral or authorize the wrong party. Fixed envelope and key derivation function (KDF) interoperability makes encrypted backups portable; hardened per-loan derivation avoids index reuse, cross-loan correlation, and expanded compromise. Those are reasons for the existing contracts, not permission to add recovery authority. Automatic organization default broadcast, institutional collateral recovery, nonresponse activation, stronger forgotten-password recovery, and complete cooperative operations remain excluded under the automatic lender-default broadcast, institutional last-resort collateral recovery, borrower-nonresponse activation recovery, stronger participant-Vault recovery, and cooperative settlement operations exclusions, each with its own revisit condition.

The architecture guide shows where this domain fits in the system, and the domain record lists exactly what the domain owns and excludes.

Other guides

This guide refers to these separate documents of the System Guide:

- How the Bitcoin-backed USDC one-loan system works (architecture guide)

Technical records

The exact rules, contracts and conditions live in these records of the design package:

- **Domain record:** Bitcoin Authorization And Recovery.
- **Responsibility records:** Bitcoin Prefund, Escrow, And Package Construction; Borrower Bitcoin Authorization And Recovery; Lender Bitcoin Authorization And Recovery.
- **Flow record:** Borrower Long-Stop Recovery.
- **Interface contracts:** Bitcoin Fee Estimate And Signed Tier Selection; Cooperative Settlement Package; Finalized X, Y, Or S To Fixed Bitcoin Action; OriginationBindingV1, TerminalBindingV1, And Package Manifests; P2A CPFP Request, Local Helper, And Bitcoin Core; Participant-Local Base Calls; Vault Envelope And Local Handoff.
- **State contracts:** Bitcoin Escrow And Terminal Outcome; Bitcoin Package Graph; Bitcoin Prefund And Origination Attempt.
- **Invariants:** Default Timing And Base Separation; Missing Y Residual Lender Risk; One Terminal Escrow Outcome; Zero-Confirmation Boundary.
- **Security boundary:** Valid repayment plus missing `Y` leaves Bitcoin unsettled.
- **Scope boundaries:** Automatic Lender-Default Broadcast; Borrower-Nonresponse Activation Recovery; Cooperative Settlement Operations And Base Accounting; Institutional Last-Resort Collateral Recovery; Stronger Participant-Vault Recovery.
- **Constraint record:** Durable Signed Exit Package Archive.